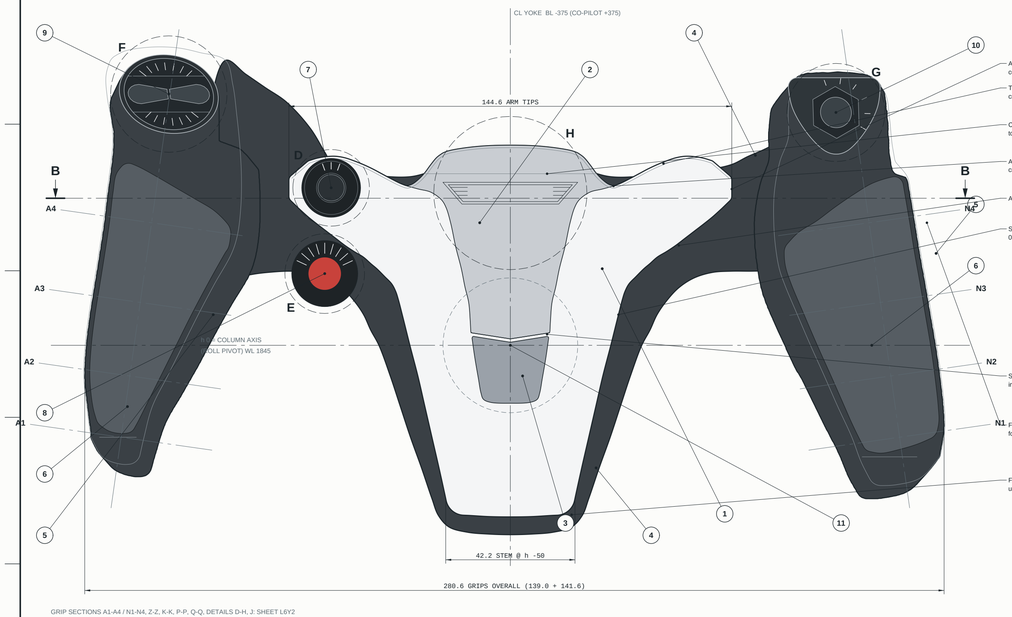
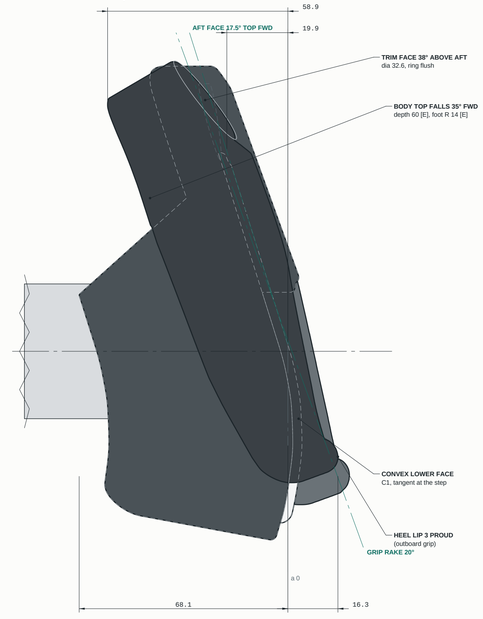


GRIP SECTIONS A1-A4 / N1-N4, Z-Z, K-K, P-P, Q-Q, DETAILS D-H, J: SHEET L6Y2

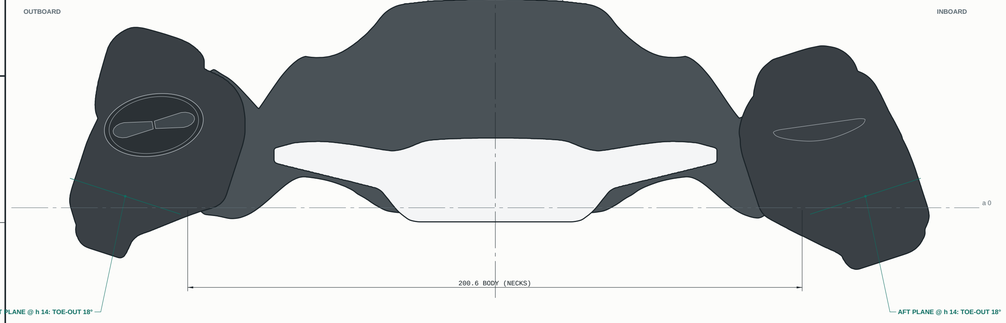
FRONT ELEVATION - PILOT YOKE

LOOKING FORWARD, AS THE PILOT SEES IT (CO-PILOT YOKE = MIRROR IMAGE) - SCALE 1:1



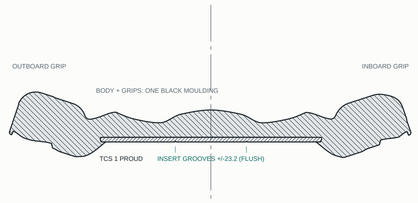
SIDE ELEVATION

FROM OUTBOARD, NOSE LEFT, NEUTRAL PITCH; a 0 = X 3700; COLUMN BROKEN (L6Y2 DETAIL J) - 1:1



VIEW FROM ABOVE

FORWARD UP, PILOT YOKE - SCALE 1:1



SECTION B-B

h 48 (BADGE / TCS), EXACT CUT OF THE SOLIDS, FROM ABOVE - SCALE 1:2

ITEMS & FINISHES (render/lookdev names; 'new' = to add)

#	PART	MATERIAL	FINISH / COLOUR [SOURCE]
1	WHITE COVER (Y SHELL)	yoke_white	gloss clear-coat white, neutral-cool (0.77 / 0.78 / 0.80), fine pearl; R 1 edges, top land [M]
2	UPPER INSERT = CROWN, BADGE	yoke_silver (new)	satin aluminium 0.62, metallic 0.85, rough 0.30; flush, 0.6 x 0.5 grooves [M]
3	LOWER INSERT	yoke_silver_dark (new)	smoke silver 0.30, metallic 0.85, rough 0.34; flush [M]
4	BLACK BODY (RIM, WEB, NECK)	grip_black	satin black, rough 0.42: crisp edge highlights; R 2 aft edge [M]
5	GRIP (2) WITH FRAME, HEEL LIP	grip_black	satin black; 2.0 frame R 1.5, heel lip 3 proud R 4 [M]
6	LEATHER PAD (2)	leather_black	charcoal 0.05, rough 0.58, 1.1 pebble grain, ~3 creases, 2.5 below the frame [M]
7	TCS BUTTON	gloss_black / black	gloss black bezel 1 proud, satin button recessed [M]
8	AFCS DISCONNECT	button_red (new) / black	red cap 0.70 / 0.035 / 0.04, clear coat 0.3; NOT emissive, on a black boss [M]
9	PITCH TRIM ON ROUND FACE	black / legend_white	satin black face, ring + rocker, tick legends 0.80 [M]
10	PUSH BUTTON ON SHIELD	black	satin black hex collar, smooth cap, centre dimple [M]
11	COLUMN	metal_dark	satin dark metal, dia 44 [E] (never seen)
12	COLUMN SURROUND (PANEL)	pedestal_gunmetal	satin dark grey J band 23 wide + dished recess [M / E]

MAIN DIMENSIONS (read from the table / the solids)

DIMENSION (mm, YOKE-LOCAL)	VALUE	SOURCE
shell height / about h 0	121.3 / -56.0, +65.3	[M] rev J landmark-pose frame
arm tips / stem at h h / at h -50	144.6 / 65.6 / 42.2	[M] sigma 1.5
crown above the arm junctions	13.3	[M] parting grooves
step / lower insert bottom	h 2.1 (V) / -18.9	[M]
black rim at h 10 / 0 / -50 / foot	13.0 / 10.1 / 4.5 / 5.8	[M] outboard true edge
aft face above the step	17.5° top fwd	[M] bundle adj., +/-4
body depth / top / cover	60 / 35° / 3	[E] / [M] / [E]
heads: outboard / inboard top	94.5 / 89.2	[M] rev I 95.4 / 95.4
grips: bottoms out / in	-42.9 / -50.1	[M] rev I -42.3
grips overall (outboard + inboard)	280.6	[M] rev I 283
rake / toe-out out, in	20° / 18°, 18°	[M] +/-5 / +/-6 (rev I 13 / 28)
grip depth bottom / mid / head	20-22 / 27 / 30	[M] / [M] / [E]
pad out: top / slot corner / bottom	59.4 / (91, 38) / -28.8	[M] diagonal top 31°
pad in: top / slot corner / bottom	55.0 / (89, 32) / -35.2	[M] diagonal top 33°
frame / leather below it / heel lip	2 / 2.5 / 3 proud	[M]
trim face centre / dia / up, turn	(111.5, 82) / 32.6 / 38°, 10°	[M] triangulated, 7 views
shield facet / button centre	31 x 25.5 / (106.3, 76)	[M] (inboard: -u)
TCS / AFCS cap / boss	(58.5, 51.4) 19.2 / 10.8 / 21.6	[M]
column dia / axis	44 [E] / h 0 = WL 1845	kept; surround reads 1821
pitch travel / roll	+/-90 [E] / +/-70 [E]	unchanged (YOKE)

KNEES vs YOKE (L6B), REV H > I > J - CRITERIA UNCHANGED

OCCUPANT	LEVEL mm	ROLL+/-70 mm	CONTACT deg	KEEP 25 deg	RUDDER mm
95th male	38 > 35 > 40	-19 > -18 > -19	17.5 > 17.5 > 15.0	5.0 > 2.5 > 5.0	-11 > -13 > -9
50th male	50 > 48 > 52	-13 > -14 > -14	25.0 > 25.0 > 22.5	10.0 > 10.0 > 10.0	5 > 3 > 7
5th male	51 > 48 > 51	-21 > -21 > -22	22.5 > 22.5 > 22.5	10.0 > 10.0 > 10.0	20 > 18 > 22
5th female	76 > 78 > 81	5 > 0 > 0	87.5 > 87.5 > 87.5	27.5 > 25.0 > 25.0	55 > 53 > 56

NOTES

1. Yoke-local mm: u outboard (pilot: -y), h up from the hub centre = the column axis / roll pivot (Wt, 1845, BL -/+375), a aft of the aft face at the insert step (X 3700 at neutral pitch). Drawn from model/interior.py YOKE_DESIGN rev J through the solids of model/yoke_geom.py; L6 / L6B take the same solids (yoke_outline_yz, yoke_sweep_pieces).
2. CURVE LAW (table curve_law): every outline is a centripetal Catmull-Rom spline through its samples, sampled every 0.25; the samples in the *_corners tables are corners (R 0 sharp, R > 0 a tangent fillet); the lower aft face is a C1 Hermite tangent to the 17.5° plane. Views are exact silhouettes of the solids, sections exact cuts (no filtering).
3. White moulding symmetric; the black body and the grips are NOT: an outboard web up to the trim head, an inboard LOW NECK with an open notch under the inboard head; switches outboard (TCS, AFCS, trim), the push button on the inboard head; the co-pilot yoke is the mirror image.
4. NO MARKINGS: the badge is an abstract winged plate, every legend a row of ticks.
5. Not modelled / unsupported: the rev H column bellows (none in 8 photos), a PTT trigger (none seen), the PC-24's extra grip tab.
6. Knee checks on the rev J solids: the 95th clears the yoke by 40 mm wings level (criterion 25: passes); roll to contact 15.0° (rev I 17.5°), keeping 25 mm to 5.0° -- owner decision (open point 1 of L6B; criteria not loosened).

OPEN POINTS FOR THE OWNER

1. Absolute fore-aft station: the locked photos put the aft face 60 +/-30 forward of the drawn neutral (hub_dx kept: the gust-lock pose?).
2. Column: the surround's joint line reads WL 1821 +/-10 (25 under the pivot, kept); fixed J surround vs a part moving with the yoke; push travel 90 [E].
3. Knees: with the rev J grips (toe 18 / 18, the wider inboard paddle) the 95th meets a grip from 15.0° of roll (rev I 17.5°): owner decision, criteria unchanged.
4. Hidden: body depth 60, the necks' depth, grip forward corners R 11, rocker / cap heights (+/-2).

REVISIONS - PC12-L6Y

REV	DATE	DESCRIPTION
A	2026-10-05	first issue (owner 2026-10-05: 'the yoke isn't quite right especially if you look at the details'): yoke re-measured on 8 PRO photos; YOKE_DESIGN rev I replaces the rev H outline
B	2026-10-05	rev J (judges r1): the yoke as solids (model/yoke_geom.py) -- every view / section exact; curve law + corner tables; one body with an outboard web and an inboard LOW NECK (open notch under the head); pads with diagonal tops in a raised 2 mm frame, rolled heel lip; grips refitted per grip (toe 18 / 18) on 4 pad views; ROUND trim face and SHIELD facet heads, triangulated switch centres; crown / arm tops 1.7 lower (landmark-pose frame), top land, crown crease; body top falling forward 35 deg; inserts flush; AFCS round boss, hex button collar, mirrored rocker lobes; sheet L6Y2 added (sections, heads, switches, column, photo check)

PILATUS PC-12 PRO

PC-24-STYLE CONTROL YOKE - VIEWS

FIRST ANGLE PROJECTION

DWG NO	REV	SCALE	SHEET	UNITS
PC12-L6Y	B	AS SHOWN	A1 · 1 OF 2	mm

DRAWN	DATE
Claude - loftkit parametric model	2026-10-05

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION